

SHASHWATA MANDAL

Ames, Iowa

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RESEARCH INTERESTS

Robotics, Path Planning, Autonomous Systems, Multi-Agent Surveillance, Computational Geometry, Reinforcement Learning, Target Tracking.

EDUCATION

Iowa State University

Ph.D. in Computer Science (GPA: 3.71)

Ames, IA

2019 – 2026 (Expected)

Pennsylvania State University

Master of Science in Computer Science (GPA: 3.47)

Harrisburg, PA

2017 – 2019

West Bengal University of Technology

Bachelor of Technology in Computer Science and Engineering (GPA: 3.41)

Kolkata, India

2012 – 2016

PUBLICATIONS

S. Mandal, S. Bhattacharya. "Scalable Multi-Agent Surveillance: A Kernel-Based Approach." *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

S. Mandal, S. Bhattacharya. "From Distributed Coverage to Multi-agent Target Tracking." *Proceedings of the International Symposium on Multi-Robot and Multi-Agent Systems (MRS)*, 2023.

S. Mandal, S. Bhattacharya. "Relay Pursuit for Multirobot Target Tracking on Tile Graphs." *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2023.

G. Laguna, **S. Mandal**, S. Bhattacharya. "Roadmap for visibility-based target tracking: Iterative construction and motion strategy." *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021.

S. Mandal, T. Gao, S. Bhattacharya. "Planning for Aerial Robot Teams for Wide-Area Biometric and Phenotypic Data Collection." *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021.

H. Saha, **S. Mandal**, et al. "Comparative performance analysis between nRF24L01+ and XBEE ZB module based wireless ad-hoc networks." *International Journal of Computer Network and Information Security*, 2017.

S.G. Dastider, H. Kashyap, **S. Mandal**, et al. "Feature subset selection for clustering using binary particle swarm optimization." *Proceedings of the International Conference on Information Technology (ICIT)*, 2015.

TEACHING EXPERIENCE

Graduate Teaching Assistant

Iowa State University

Ames, IA

2019 – Present

- **COMS 511**: Design and Analysis of Algorithms (Graduate Level)
- **COMS 311**: Introduction to the Design and Analysis of Algorithms
- **COMS 321**: Computer Architecture
- **DS 2010**: Introduction to Data Science
- Facilitated lab sessions, graded assignments/exams, and held office hours for classes of 50+ students.

RESEARCH EXPERIENCE

Research Intern

Army Research Lab

March 2025 – Present

Adelphi, MD

- Designing an entropy-based controller to maintain line-of-sight tracking of an intruder in a 2D map environment.
- Developing a Corner-Aware Risk map to identify and predict escape routes for intruders.
- Creating an evaluation metric to select optimal exploration parameters (alpha) in unsupervised learning based on empowerment signals.

Visual Target Tracking with NeRF | Iowa State University

Oct 2025 – Present

- Investigating the utility of Neural Radiance Fields (NeRF) to approximate geometric properties for indoor navigation.

- Developing methods to leverage pre-trained models for efficient visibility estimation in cluttered environments.
- Exploring techniques to accelerate line-of-sight computations for real-time tracking applications.

Scalable Multi-Agent Surveillance | *Iowa State University*

Aug 2023 – Dec 2024

- Developed a unified control framework that generalizes distributed coverage algorithms to solve multi-target tracking problems.
- Utilized distributed geometric kernels to partition non-convex environments into star-shaped visibility regions, ensuring agents maintain line-of-sight connectivity.
- Deriving decentralized control laws that ensure scalability, maintaining constant computational complexity regardless of the number of agents.

Battery Management System for Last-Mile Delivery | *Iowa State University*

Jan 2024 – Present

- Designing a drone-assisted BMS for on-demand battery swaps to ensure continuous vehicle operation.
- Developing algorithms for optimal battery health and efficiency, implementing an RL-based controller for operation in uncertain environments.

Multi-Robot Relay Pursuit for Target Tracking | *Iowa State University*

Jan 2022 – May 2023

- Designed a handover strategy (Relay Pursuit) for multiple robots to track an intelligent target on tile graphs.
- Formulated the problem to minimize the number of robots required while guaranteeing capture.
- Published results in ICRA 2023 demonstrating efficiency over traditional pursuit-evasion methods.

Aerial Swarms for Agricultural Data Collection | *Iowa State University*

Aug 2020 – Aug 2021

- Developed path-planning algorithms for aerial robot teams to collect wide-area phenotypic and biometric data.
- Optimized trajectories for drone swarms to ensure efficient coverage of large-scale agricultural fields.
- Co-authored paper published in IROS 2021 detailing the visibility-based tracking and data collection framework.

INDUSTRY EXPERIENCE

Associate Software Engineer

Aug 2016 – Aug 2017

Nomura Research Institute Financial Technologies Ltd.

Kolkata, India

- Built and maintained frontend/backend features for a financial web platform using Java, Spring, and JQuery.
- Managed release cycles and supported team coordination during critical delivery phases in an Agile environment.

TECHNICAL SKILLS

Languages: C++, Python, C, Java, Rust, Arduino

Robotics & AI: ROS/ROS2, Gazebo, Path Planning, Multi-Agent Systems, Reinforcement Learning, SLAM, PyTorch, OpenCV, NeRF.

Tools: Git, Docker, Linux, LaTeX, Spring Framework.

HONORS & AWARDS

Travel Grant | *IEEE/RSJ IROS*

2021

Graduate Research Assistantship | *Iowa State University*

2019 – Present